1. In general, the back button on projects media carousels does not work. Needs to be addressed sitewide
2. Add option to download in top right when media is clicked and opened in its detailed carousel view
3. Preview images of videos before they're clicked on should not be so poor quality
4. Remove all em dashes
5. When i say to set an image to the Xth in the carousel, i also mean to shift the following ones down 1 slide
6. If i say to change the order of the Projects themselves, i mean how they appear in their subcategory on both their category page, as well as the broad Projects page
7. Remember that the original high quality files of nearly all media for the projects can be found in C:\Users\Andre\Downloads\1. Projects-20260929T110050Z-1-001
8.

Robotics

Computer Vision & Perception

- Robust Visual Navigation Using Semantic Segmentation and Sensor Fusion
    - Set The perception pipeline and pedestrian interaction running live. As main first item carousel,
    - “The polynomial-fit centreline and the navigation waypoint handed to ROS2.” should be second item, and look on media to replace this image with a better quality version from file explorer
    - “Calibrating the Logitech BRIO against the Ouster OS0-64 with a checkerboard target.” should be third
    - Leave as Pending Task on why FullDeliveryDemo and TrailborClimbingHill are low resolution
    - Change caption from “TRAILbot navigating a hiking trail during field trials.” to “TRAILbot navigating a hiking trail, identifying hiker, and using voice interaction to offer snacks”
    - Change caption from “The perception pipeline and pedestrian interaction running live.” to a better description of the 4 quadrants in the FullMappingAndTraversal1 video. FLAG FOR REVIEW
    - Switch with better quality image for “The system's state machine, from trail following through to human interaction.”
    - Rewrite to remove the em dash in the text
    - Use TRAIL lab link https://www.trailab.utias.utoronto.ca/
    - The “see here for more details” should link to https://andrescerverarozo.au/real-time-trail-segmentation-model-for-an-autonomous-robot/
- YOLO-Driven 3D Traffic Mapping from Stereo Camera

    - For just this project, on desktop version only, vertically align the three images above each other
- Georeferenced Ground Targets Using UAV Imagery
    - In the detailed project page, move the image of the drone “The Parrot AR.Drone 2.0 that captured the downward-facing imagery.” to 5th place in carousel
    - Set “Noisy per-frame detections consolidated into final target estimates by k-means.” to be third image in carousel
    -
- 3D Point Cloud Matching Using SVD and ICP
    - For only this project, on desktop only, set the 2 images to be the same size and next to each other

Drones & Aerial Robotics

- Dynamic Modeling and Control of a UAV in Simulink
    - FLAG AND NOTIFY ME FOR MAJOR REWRITE
- Sim to Real Drone Racing with RRT* Planning
    - Flag to investigate video quality of video
    - Replace video caption with “Flight footage of the Crazyflie running a planned trajectory through an obstacle course.”
    - Replace with better image quality for “The planned path in PyBullet, weaving through the gates in the order revealed at runtime.”

Localization

- Maze-Navigating Robot Localization Using 2D LiDAR and Particle Filter SLAM
    - Set image “RViz during a run: occupancy map, particle cloud and the planned path.” to be second in the carousel
    - Keep image “Hector SLAM builds the map under teleop, then AMCL localizes against it during the run.” as third in carousel
    - FLAG FOR ME TO CHECK FOR VERTICAL HEIGHT OF PAGE AFTER CHANGES
    - Link in text should go here https://andrescerverarozo.au/embedded-omni-directional-maze-navigating-robot-with-360-lidar/
    -
- EKF-Based Probabilistic Localization Using Sensor Fusion
    - Set “The Vicon room, whose ground truth was used to derive the sensor variances.” photo to be 5th in carousel
    - Set “The nonlinear motion and landmark observation models the EKF linearizes around.” to be second photo in carousel
    - Change video caption replace “filter” with “Extended Kalman Filter localization”
    - Remove Drones tag from this project and add Python
- Drift-Resilient Visual Odometry with Stereo Vision
    - Set this project to be the third one displayed between the 3 options of the Localization projects
    - Change caption of vid to something that explains that the yellow lines moving are the detected changes between frames of detected features

Motion Planning

- Adaptive Heuristic Exploration for Autonomous Mapping on Turtlebot
    - This applies for ALL PROJECTS, is there a way for the preview image on videos to not be such poor image resolution?
    - Set “The simulated environment used before hardware runs.” image to be second in carousel
- Manipulator Trajectory Optimization Using Interior-Point Methods
    - The first and last gif of the carousel is duplicated, remove the last, then for the first gif, add in the description that it is for a 2-link manipulator
    - Set “The end-effector path with obstacle avoidance disabled, then enabled.” image as second in carousel
- Sampling-Based Motion Planning: Speed vs. Optimality
    - Set “RRT* explores far more of the space, and converges on a 15.37m path.” as first image in carousel

Programming

Data Science

- Geolocating Fuel Stations Using Vehicle Telematics and BigQuery
    - Set project to be the 1st project listed under “Data Science” projects
    - Set “Detected clusters checked against satellite imagery - the final filter against false positives like depots.” as first image of carousel
    - Keep San Francisco as second image of carousel
    - Set “Fuel level traces for ten vehicles over three days - the sudden jumps DBSCAN clusters into fill events.” as third image
    - Rewrite the captions of carousel to make sense with new ordering and flag for me to review manually after
    -
- Geospatial Signal Strength Analysis Using BigQuery
    - Set project to be the 2nd project listed under “Data Science” projects
    -
- Automated Cloud Pipeline for Antarctic Environmental Sensor Data
    - Set project to be the 3rd project listed under “Data Science” projects
    - Set “All three deployment units, each independently logging and reporting its own environmental data.” to first item of carousel
    - Set “How the GR9 telematics device and its USB sensors - thermometer, compass, weather station, light sensor - wire into the logger.” as second image
    - Set “The automated cloud pipeline for the Antarctic sensor logs, from starred email to the BigQuery antarctica_logs table.” as third image
    -
    -
- Human Activity Classification Using Azure AutoML
    - Set project to be the 4th project listed under “Data Science” projects

    - Move “Model evaluation on the test set: ROC, precision-recall, calibration and confusion matrix.” image to second and keep “The Azure Machine Learning workflow, from data preparation through to deployment.” as third
    - Improve image quality on all images in this project if possible with replacements from File Explorer
    -

Machine Learning

- Transformer-Based English-to-French Translation
    - Set project to be the 1st project listed under “Machine Learning” projects
- Real-Time Trail Segmentation Model for An Autonomous Robot
    - Set project to be the 2nd project listed under “Machine Learning” projects
    - Set “The LEDNet encoder-decoder architecture.” as 3rd image of carousel
    - The link in the text should redirect to https://andrescerverarozo.au/robust-visual-navigation-using-semantic-segmentation-and-sensor-fusion/
    -
- Reinforcement Learning: Neural Policy Optimization in CartPole
    - Set project to be the 3rd project listed under “Machine Learning” projects
- Multi-Class Animal Classification with CNNs
    - Set project to be the 4th project listed under “Machine Learning” projects
    - Set “The network architecture, from RGB input through to the class prediction.” to be first image in carousel
    -

Electrical/Embedded

Embedded Systems & Prototyping

- Gesture-Controlled Robot with Raspberry Pi and Arduino Integration
    - Set project to be the 1st project listed under “Embedded Systems & Prototyping” projects
    - Set “The finished robot, built to respond to hand gestures with lights, a horn, or autonomous navigation.” as the 2nd item in the carousel
    - Set “OpenCV colour tracking running live, one of the autonomous behaviours a gesture could trigger.” as 3rd item in carousel
- Embedded Omni-Directional Maze-Navigating Robot with 360° LiDAR
    - Set project to be the 2nd project listed under “Embedded Systems & Prototyping” projects
    - Link in text should redirect to https://andrescerverarozo.au/maze-navigating-robot-localization-using-2d-lidar-and-particle-filter-slam/
    - Set “The robot retrieving a block in the checkered maze.” as first image in carousel

    - Set “What sits on each layer, from the lidar down to the motor controllers.” as second image in carousel
    - Set “The full wiring diagram and component list.” as third image in carousel
- Microcontroller-Driven Beverage Automation Platform
    - Set project to be the 3rd project listed under “Embedded Systems & Prototyping” projects
- U of T Design League Winner – Secure Embedded Vaccine Transport
    - Set project to be the 4th project listed under “Embedded Systems & Prototyping” projects
    - Set “The case open, showing the vial rack alongside the keypad and display.” as 2nd image in carousel

Hardware & Power Systems

- Battery Characterization and PCB Assembly for High-Voltage Electric Vehicle
    - Set project to be the 1st project listed under “Hardware & Power Systems” projects
    - Set “The project poster: a safe, efficient power architecture for a Hyperloop pod.” as first item of carousel
- Oscilloscope Analog Interface and PCB Design
    - Set project to be the 2nd project listed under “Hardware & Power Systems” projects
    - Replace images with better quality copies from file manager
- Real-Time Monitoring of eMMC Health in IoT Devices
    - Set project to be the 3rd project listed under “Hardware & Power Systems” projects
    - Set “How erase counts are distributed across blocks, and the cumulative spread.” as first image in carosuel
    -

Mechanical

- SolidWorks Design of a Home CNC Mill
    - Set project to be the 1st project listed under “Mechanical” projects
    - Set “The gantry carrying the spindle across the X axis.” as first image of carousel
    - Set “Detail drawing of the spindle mechanism that moves the cutting tool through the work area.” as second image
    - Set “The complete mill, sized and priced for personal-scale use rather than an industrial shop.” as third image of carousel
    - Report button should take user to https://drive.google.com/file/d/1zKQvUtiTDM--oEFIVEU9KSUrDFHfJps8/view?usp=sharing
    - Rewrite to remove all em dashes
- Stress and Finite Element Analysis (FEA) of an Aircraft Stringer
    - Set project to be the 2nd project listed under “Mechanical” projects

    - Set “Photoelastic fringe orders at 20% load, used to read stress at the critical points.” as 3rd image of carousel
    - Change text in button from “Report” to “Reports”
- Dynamic Force Analysis and Redesign of an Outdoor Rowing Machine
    - Set project to be the 3rd project listed under “Mechanical” projects
    - Turn the drawings images to landscape
    - Set “Seat acceleration, velocity and handle force through a full stroke.” image to be third image in carousel
- CAD Reconstruction of Apple MacBook Pro
    - Set project to be the 4th project listed under “Mechanical” projects
    - Set “Front-on view, checking the display and chassis proportions against the real 15-inch MacBook Pro.” as first image of carousel
    - Set “Close-up of the keyboard, where individual key lettering was modelled rather than approximated.” as third image of carousel
- Heliostat Optimization and Solar Tower Design in Energy3D
    - Set project to be the 5th project listed under “Mechanical” projects
    -
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